MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, XM684

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 40MM, HE, XM684.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-P-60942	-	Primer Percussion for 40MM Ammunition

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions

FSC: 1310

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MIL-STD-331 - Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-1168 - Lot Number of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9247850 - Cartridge, 40MM, HE, XM684

PUBLICATIONS

ARMY

ET-9247850 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.



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3.2 Cartridge.-The cartridge shall comply with all requirements specified on drawing (dwg) 9247850, all associated drawings and with all requirements specified in applicable specifications.

3.3 X-ray examination of fuze assemblies prior to assembling to body assembly.-The fuze assembly shall be x-rayed for improperly assembled, missing parts, armed or partially armed (as evidenced by the detonator moving any degree from the safe position).

3.4 Air pressure.-The cartridge shall withstand an air pressure of three (3) pounds per square inch (psi) for five (5) seconds, minimum (min) without a drop in pressure when tested as specified in 4.4.8.

3.5 Functioning

3.5.1 Cartridge.-The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 790 $\pm$ 15.0 feet per second (fps) and a standard deviation not exceeding 15.0 fps when tested as specified in 4.4.9. The action time of the cartridge when tested as specified in 4.4.9 shall be 4.0 milliseconds, maximum.

3.5.2 Safety.-There shall be no premature burst (see 6.7) or evidence thereof in any of the test firings when tested as specified in 4.4.9.

3.6 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein,

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unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-Not applicable.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used throughout this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to

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exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Cartridge case assembly metal parts from one lot interfix number from one manufacturer.

b. Metal parts from one lot interfix number from one manufacturer.

c. Fuzes from no more than two (2) consecutively produced serial lots from one manufacturer.

d. Primers from one lot number from one manufacturer.

e. Propellant from not more than one (1) lot number from one manufacturer.

f. Composition A5 from one lot interfix number from one manufacturer.

g. Cartridges in less than full belt quantities from any acceptable lot of cartridges shall be held until a cumulative amount of 3000 cartridges are obtained or until 1 year has elapsed. These cartridges shall be linked, packed and shipped in multiples of 50 as a conglomerate lot. No further ballistic firing need be performed on these cartridges, since these cartridges came from acceptable lots. Acceptable lots are not accepted on waiver. If at the end of 1 year, 3000 cartridges are not obtained, the cumulative amount of cartridges shall be linked, packed and shipped in multiples of 50 and the remaining cartridges shall be destroyed.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect

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to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cartridge case assembly, prior to loading (see dwg. 8886327 covering a detail of dwg. 9247850)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Closing cup missing	Visual	01001
Minor:	AQL 0.65 Percent		
201.	Talc assembly missing	Visual	01002

4.3.2.2 Cartridge case assembly, prior to inserting base plug and primer (see dwg. 8886327 covering a detail of dwg. 9247850)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Spilled powder from cartridge case	Visual	02001
2.	Propellant charge weight obviously incorrect (see 6.9)	Visual	02002
Major:	100% Inspection		
101.	Propellant charge weight obviously incorrect (see 6.9)	Visual	02003
Minor:	None defined		

4.3.2.3 Cartridge case assembly (see dwg. 8886327 covering a detail of dwg. 9247850)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Primer above flush with base plug	Gage	03001

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Major:	AQL 1.00 Percent		
101.	Base plug above flush	Gage	03002
102.	Depth to base plug, max.	Gage	03003
103.	Depth to primer, max from base plug	Gage	03004
104.	Primer damaged	Visual	03005
105.	Excessive sealing compound on exterior surface of primer	Visual	03006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	03007

4.3.2.4 Body loading assembly (see dwg. 8886400 covering a detail of dwg. 9247851)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Depth to charge	Gage	04001
102.	Diameter of rotating bands	Gage	04002
Minor:	AQL 1.50 Percent		
201.	Threads damaged	Visual	04003
202.	Evidence of poor workmanship (see 3.6)	Visual	04004
203.	Charge parallel to rim of body	Gage	04005

4.3.2.5 Projectile, prior to crimping cartridge case assembly (see dwg. 9247850)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 Percent		
201.	O-ring (packing) missing in groove	Visual	05001

4.3.2.6 Cartridge (see dwg. 9247850)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.65 Percent		
101.	Gap, min.	Gage	06001
102.	O-ring exposed	Visual	06002
103.	Gap, rear rotating bands	Gage	06003
Minor:	AQL 1.50 Percent		
201.	Excessive adhesive on exterior surfaces	Visual	06004
202.	Marking missing	Visual	06005
203.	Evidence of poor workmanship (see 3.6)	Visual	06006

4.3.2.7 Linked ammunition, prior to packing (See dwg. 8886405)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Link improper (anyone of the six (6) tabs of link not in groove; link inverted; link damaged; link distorted	Visual	07001
Major:	None defined		
Minor:	None defined		

4.3.2.8 Unsealed fiber box (see dwg. 8865432)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Cartridge inverted	Visual	08001
102.	Incorrect number of cartridges	Visual	08002
103.	Fillers missing	Visual	08003
Minor:	None defined		

4.3.2.9 Sealed fiber box, prior to sealing bag (see dwg. 8865432)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 Percent		
201.	Container damaged	Visual	09001
202.	Corners not blunted	Visual	09002
Minor:	AQL 1.00 Percent		
201.	Contents loose	Manual	09003
202.	Label missing or unidentifiable	Visual	09004

4.3.2.10 *Unsealed wooden packing box (see dwg. 8865431)*

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Marking of bag misleading or unidentifiable	Visual	10001
102.	Bag torn or perforated	Visual	10002
103.	Bag improperly sealed	Visual	10003
Minor:	None defined		

4.3.2.11 Sealed wooden packing box (see dwg. 8865431)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined—		
Major:	AQL 0.40 Percent		
101.	Box damaged	Visual	11001
Minor:	AQL 1.50 Percent		
201.	Contents loose	Manual	11002
202.	Car seal missing or improperly positioned	Visual	11003
203.	Marking misleading or unidentifiable	Visual	11004

4.3.3 Testing

4.3.3.1 Check test for deterioration of primers (See MIL-P-60942) Major Defect, Code No. 12001.-If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridge. (See 6.4)

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4.3.3.2 Propellant weight of case, cartridge (See dwg. 8886327) Major Defect, Code No. 13001.-The propellant weight shall be determined and then check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on he applicable drawings shall be classified defective and removed from the lot.

4.3.3.3 Security of projectile (see dwg. 9247850) Major Defect, Code No. 14001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter J, single plan, with an AQL of 0.65 percent. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

4.3.3.4 Security of crimp of base plug on cartridge case assembly (see dwg. 8886327) and action time of cartridge case assembly (see 3.5.1) Major Defect, Code No. 15001.-Five (5) primed base plug assemblies shall be randomly selected every four (4) hours from each manual primer assembly machine or twenty (20) base plug and primer assemblies shall be randomly selected every four (4) hours from each automatic primer assembly. These assemblies shall be assembled to cartridge cases for test of "Security of crimp of base plug and action time" determination. The base plug and primer assemblies produced by each primer assembly machine shall be kept segregated and identified. They shall not be used in production until the above tests have been performed. If any base plug moves more than .005 above flush with rear of case, or the action time of any cartridge case assembly exceeds 4.0 milliseconds, the four (4) hours production from each manual primer assembly or each automatic primer assembly machine involved shall be rejected. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Specific gravity of the charge of body loading assembly (see dwg. 8886400) Major Defect, Code No. 16001.-Two (2) sample cup assemblies shall be selected from each four (4) hour's production and subjected to this test. If any sample fails to comply with the specific gravity requirement specified on the applicable drawing, the four (4) hour's production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.3.

4.3.3.6 Security of fuze on projectile assembly (see dwg. 9247851) Major Defect, Code no. 17001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. Samples shall be randomly selected after assembly and the test performed before adhesive hardens. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

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4.3.3.7 Total length of cartridge (see dwg. 9247850) Major Defect, Code No. 18001.-Each cartridge shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.8 Chamber gage failure (see dwg. 9247850) Major Defect, Code No. 19001.-Each cartridge shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.6 using equipmet in accordance with 4.3.4.

4.3.3.9 Air pressure (see para. 3.4) Major Defect, Code No. 20001.-The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter P, single plan, with an AQL of 0.015 percent. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.10 X-ray examination of fuze assemblies prior to assembling to body assembly (See 3.3), Code No. 21001.-Each fuze assembly, prior to assembling to body assembly shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled, having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.7.

4.3.3.11 Functioning (see para. 3.5), Code No. 22001.-The Government Inspector shall select one-hundred (100) cartridges from each lot and subjected to the test specified in 4.4.9. Samples shall be randomly selected prior to linking.

4.3.3.11.1 Rejection.-The lot shall be rejected if any of the following occur:

a. Critical.-Any premature burst occurs (see 6.7).
b. Critical.-Any primer blows back (as evidenced by perforation of the primer cup or smoke deposit on base plug or cartridge case base).
c. Critical.-Any projectile sticks in the gun bore.
d. Major.-Mean velocity and/or standard deviation fail to meet the applicable requirement.
e. Major.-Two (2) or more primers misfire.
f. Major.-Eighteen (18) or more fail high order airburst detonation at target area (see 6.12).
g. Major.-Total of thirteen (13) or more fail high order airburst detonation plus high order detonation at impact.
h. Major.-Three (3) or more base plugs are more than .005 above flush with rear of cartridge case (see 4.4.10) or one (1) or more base plugs exceed .010 with rear or cargridge case or jams the automatic weapon.

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i. Major.-The action time of one (1) or two (2) cartridges is over four (4) milliseconds with no cartridges exceeding ten (10) milliseconds, an additional 200 cartridges shall be tested. The results for the 325 cartridges shall be combined. If three (3) or more cartridges are over four (4) milliseconds or if one (1) cartridge is over ten (10) milliseconds, the lot shall be rejected.

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.3.3.11.2 Suitability test plan (Information test only) Code No. 21002.-The first lot produced by each producer, an extra sample of ninety (90) rounds shall be selected from that lot, shipped to a government proving ground , pre-conditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds +140 degrees F. $\pm$ 5 degrees F.
30 round ambient (between +40 and +110 degrees F.)
30 rounds -40 degrees F. $\pm$ 5 degrees F.

d. Fire the rounds in accordance with paragraph 4.4.9 and with TECOM Material Test Procedures, TECP 700-700 Volume 4. Test results, failure analysis reports, and other information as requested, shall be forwarded to Picatinny Arsenal, ATTN: SMUPA-ND3 for evaluation.

4.3.3.11.3 Test validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

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4.3.3.12 Heat seal test of packing seals (see dwg. 8865432) Major Defect, Code No. 23001.-The sampling for this test shall be selected in accordance with MIL-P-116. Rejection shall be in accordance with MIL-P-116. The test shall be performed as specified in 4.4.11.

4.3.3.13 Rapid fire functioning, Major Defect, Code No. 24001.-One hundred (100) cartridges shall be selected from a conglomorated lot (M684 and M384 linked) and subjected to this test. If any gun stoppage occurs or if any projectile sticks in the gun bore, the lot shall be rejected. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9247850 and Equipment Lists (EL's) referenced theron identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Security of crimp of base plug on cartridge case assembly.-The cartridge case assembly shall be assembled to a projectile and fired. Measurement for base plug more than .005 above flush from the rear of the cartridge case shall be measured from the highest point radially. Observation shall be made for action time using equipment specified on the Equipment List.

4.4.2 Security of projectile.-The cartridge shall be placed in the fixture shown on the Equipment List and the axial tension specified on dwg. 9247850 shall be applied. Cartridge case assemblies shall not be reused. Projectile assemblies may be reused. The cartridge shall be pulled until separation occurs and the data recorded.

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4.4.3 Specific gravity of the charge of body loading assembly.- The specific gravity of the charge shall be determined as follows:

Weight the empty body assembly in grams and record as We. Fill the empty body assembly with water at 75 degrees ± 2 degrees F., weigh in grams and record the weight as Ww. Determine the volume by subtracting We from Ww. Load the assembly with charge, weigh in grams and record the weight as Wl. Determine the charge weight by subtracting We from Wl. Calculate the specific gravity as follows:

$$\text{Specific Gravity} = \frac{Wl - We}{Ww - We - 7.27}$$

4.4.4 Security of fuze on projectile assembly.-The projectile assembly shall be placed in the fixture shown on the Equipment List and the minimum torque shall be applied. Observation shall be made for the requirement specified on Dwg. 9247851. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Total length of cartridge.-Using the equipment specified on the Equipment List, the cartridge must meet the requirement specified on Dwg. 9247850. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.6 Chamber gage failure.-The cartridge shall be placed in the chamber shown on the Equipment List and must meet the requirement specified on Dwg. 9247850. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.7 X-ray examination of fuze assembly, prior to assembling to body assembly.-Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by x-ray equipment or any other method satisfactory to the Contracting Officer. This test shall be conducted in two (2) planes. From the top of the fuze assembly down and from the side.

4.4.8 Air pressure.-The cartridge shall be placed in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Functioning.-This test shall be conducted at a Government owned proving ground. The rounds being tested, including the standard control rounds, shall be temperature conditioned at 70°F ± 5°F for at least eight (8) hours, then fired within fifteen (15) minutes after being removed from temperature soak. The test and standard control rounds shall be fired at a rate not to exceed one round per minute

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using an XM129 launcher as specified on the Equipment List. The XM129 launcher shall have been "broken-in" if new, with at least forty (40) rounds and placed on an approved mount for running the test. At least 6 gun-warmers shall be rapid fired at the start of the test. After the gun-warmers have been fired, twelve (12) of the standard control rounds shall be fired. These shall be followed by the test samples and then twelve (12) more of the standard control rounds. Distance to impact area shall be 600 to 1200 meters. Impact media shall be a gravel pad (50 ft wide by 100 ft long, min.). The angle of firing shall be adjusted to give an approach angle of approximately 15 degrees. If a round falls outside the specified impact area it shall be declared a "no-test" round. The velocity shall be measured at 30 feet from the muzzle of the launcher and check measured at the same base line. The monitoring devices and time recorders shall be a dual system set-up for recording two (2) separate readings and shall be within 3 feet per second. If the two (2) readings differ more than 3 feet per second (individual shots), the results shall be discarded and another round fired in its place. When proving ground reports results for functioning, the number of all "no-test" rounds shall also be reported. "No-test" rounds shall be reported as "no-test" rounds outside the impact area or "no-test" rounds for velocity readings. Observations shall be made for functioning and results calculated and recorded for (a) cartridge action time, using the equipment specified on the Equipment List and (b) mean velocity and standard deviation (See 6.6). The mean velocity of the lot being tested shall be corrected by the correction factor determined for the particular test launcher being used on the day of the test firing. When velocity values are suspected to be outliers, they shall be tested using the statistical procedure approved by the Government proving ground and the mean velocity and standard deviation adjusted accordingly. However, no more than two (2) on the high side and no more than (2) on the low side shall be considered for exclusion. The correction factor for the test launcher shall be determined by substracting the mean velocity of the standard control rounds fired during the test from the standard velocity previously established for the standard control rounds. Either a positive or a negative correction factor will result depending on the mean velocity of the standard control rounds fired during the test. Algebraically add the resultant correction factor to the mean velocity of the test rounds.

4.4.10 Movement of base plug.-The movement of the base plug shall be measured from the highest point on the base plug to a point on cartridge case directly beside it, radially.

4.4.11 Heat seal test of packing seals.-This test shall be conducted in accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

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4.4.12 Rapid fire functioning.-This test shall be conducted at a Government owned proving ground. The cartridge shall be fired from the XM129 launcher specified on the Equipment List in burst of approximately ten (10) rounds. Observation shall be made for gun stoppage and projectile stuck in the gun bore.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with Dwg. 8865432.

5.2 Packing

5.2.1 Level A.-The cartridge preserved and packaged in accordance with 5.1.1, shall be packed in accordance with Dwg. 8865431.

5.3 Marking.-Marking shall be in accordance with Dwgs. 8796522 and Code of Federal Regulations, Title 49 Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraphs 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detaild drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finsihes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted

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contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdication of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

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Where:

X = Each individual value
$\bar{X}$ = Sample arithmetic mean $\frac{\Sigma X}{n}$
n = Sample size
$(X - \bar{X})^2$ = the sum of the squares of the differences between the sample mean and each individual values.
$n - 1$ = one less than the sample.

6.6.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used:

$$S = \sqrt{\frac{n \Sigma X^2 - (\Sigma X)^2}{n (n-1)}}$$

6.7 Premature burst.-A premature burst is considered to have occurred if the round functions in gun bore or less than 90 meters. Monitored by a spectrum analyzer as shown on the Equipment List.

6.8 Combining of proving ground tests.-When the contractor for the cartridge is also the contractor for one or more of the component thereof, the proving ground test of the contractor may be combined with the proving ground test of the cartridge, to save expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specifications shall be conducted.

6.9 If any cartridge case, prior to insertion of the base plug and primer, is suspected of containing an underweight or overweight propellant charge, based on visual inspection, it will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75 percent or more than 125 percent of the assessed propellant lead wire be classified as a critical defect and removed from the lot. Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant load will be classified as a major defect and removed from the lot.

6.10 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

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6.11 Proving ground test summary:

Test	Sample Size	Requirements
Functioning	100	See 3.5

6.12 Proper ambient functions are defined as from above the surface to 25 feet.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1310-A-018

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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